MATLAB Examples

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By MathWorks

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Econometrics Toolbox

Copy and run in your MATLAB:

```
openExample('econ/GenerateVECModelImpulseResponsesExample')
```

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Generate VEC Model Impulse Responses

This example shows how to generate impulse responses from this VEC(3) model (docid:econ_ug.brz_lcd, Ch. 6.7):

$$\begin{aligned}\Delta y_t &= \begin{bmatrix} 0.24 & -0.08 \\ 0 & -0.31 \end{bmatrix} \Delta y_{t-1} + \begin{bmatrix} 0 & -0.13 \\ 0 & -0.37 \end{bmatrix} \Delta y_{t-2} + \begin{bmatrix} 0.20 & -0.06 \\ 0 & -0.34 \end{bmatrix} \Delta y_{t-3} \\ &+ \begin{bmatrix} -0.07 \\ 0.17 \end{bmatrix} \begin{bmatrix} 1 & -4 \end{bmatrix} y_{t-1} + \varepsilon_t\end{aligned}$$

y_t is a 2 dimensional time series. $\Delta y_t = y_t - y_{t-1}$. ε_t is a 2 dimensional series of mean zero, Gaussian innovations with covariance matrix

$$\Sigma = 10^{-5} \begin{bmatrix} 2.61 & -0.15 \\ -0.15 & 2.31 \end{bmatrix}.$$

Specify the VEC(3) model autoregressive coefficient matrices B_1, B_2, and B_3, the error-correction coefficient matrix C, and the innovations covariance matrix Σ.

```
B1    = [0.24 -0.08;
         0.00 -0.31];
B2    = [0.00 -0.13;
         0.00 -0.37];
B3    = [0.20 -0.06;
         0.00 -0.34];
C     = [-0.07; 0.17]*[1 -4];
Sigma = [ 2.61 -0.15;
         -0.15  2.31]*1e-5;
```

Compute the autoregressive coefficient matrices that compose the VAR(4) model that is equivalent to the VEC(3) model.

```
B = {B1; B2; B3};
A = vec2var(B,C);
```

`A` is a 4-by-1 cell vector containing the 2-by-2, VAR(4) model autoregressive coefficient matrices. Cell `A{j}` contains the coefficient matrix for lag `j` in difference-equation notation. The VAR(4) is in terms of y_t rather than Δy_t.

Compute the forecast error impulse responses (FEIR) for the VAR(4) representation. That is, accept the default identity matrix for the innovations covariance. Specify to return the impulse responses for the first 20 periods.

```
numObs = 20;
IR = cell(2,1); % Preallocation
IR{1} = armairf(A,[],'NumObs',numObs);
```

To compute impulse responses, `armairf` filters an innovation standard deviation shock from one series to itself and all other series. In this case, the magnitude of the shock is 1 for each series.

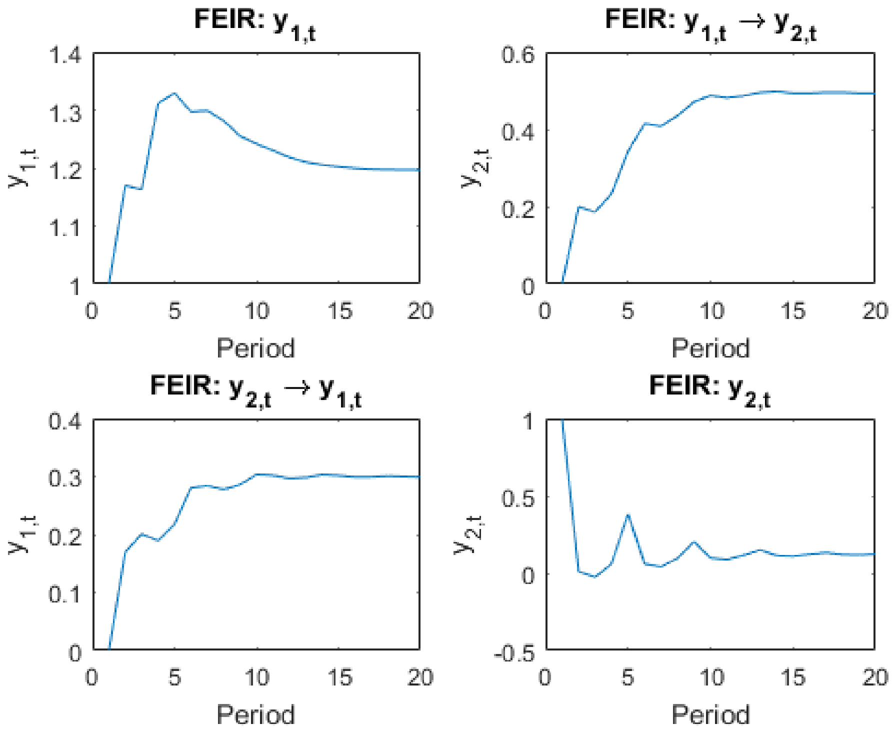
Compute orthogonalized impulse responses by supplying the innovations covariance matrix. Specify to return the impulse responses for the first 20 periods.

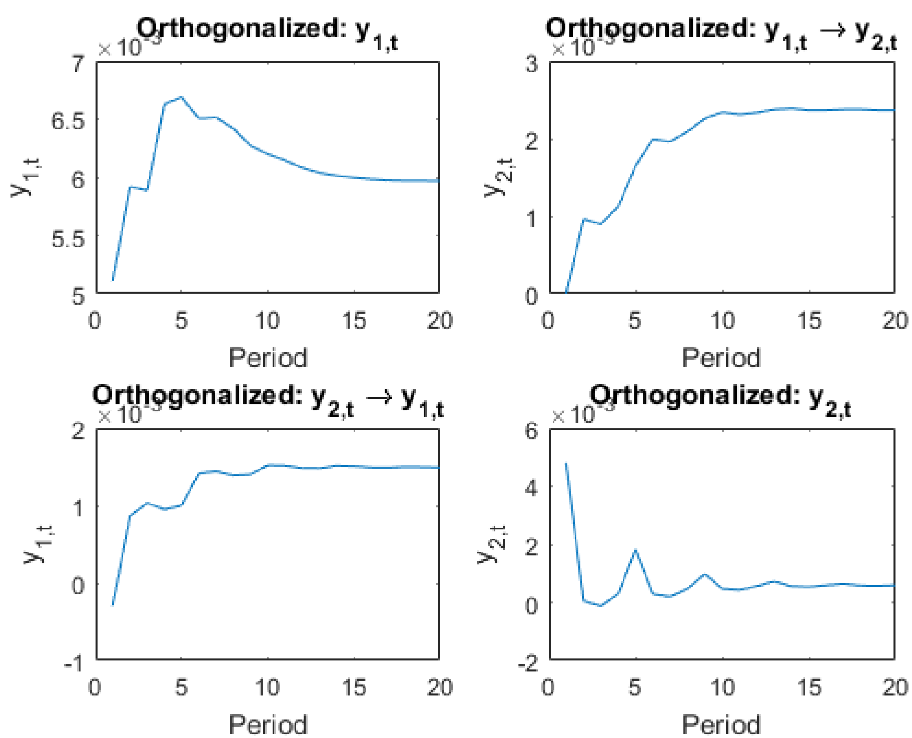
```
IR{2} = armairf(A,[],'InnovCov',Sigma,'NumObs',numObs);
```

For orthogonalized impulse responses, the innovations covariance governs the magnitude of the filtered shock.

Plot the FEIR and the orthogonalized impulse responses for all series.

```
type = {'FEIR','Orthogonalized'};
for j = 1:2;
    figure;
    imp = IR{j};
    subplot(2,2,1);
    plot(imp(:,1,1))
    title(sprintf('%s: y_{1,t}',type{j}));
    ylabel('y_{1,t}');
    xlabel('Period');
    subplot(2,2,2);
    plot(imp(:,1,2))
    title(sprintf('%s: y_{1,t} \\rightarrow y_{2,t}',type{j}));
    ylabel('y_{2,t}');
    xlabel('Period');
    subplot(2,2,3);
    plot(imp(:,2,1))
    title(sprintf('%s: y_{2,t} \\rightarrow y_{1,t}',type{j}));
    ylabel('y_{1,t}');
    xlabel('Period');
    subplot(2,2,4);
    plot(imp(:,2,2))
    title(sprintf('%s: y_{2,t}',type{j}));
    ylabel('y_{2,t}');
    xlabel('Period');
end
```





Because the innovations covariance is almost diagonal, the FEIR and orthogonalized impulse responses have similar dynamic behaviors (docid:econ_ug.brz_lcd, Ch. 6.7). However, the scale of the plots are markedly different.

Related Examples

Simulate Responses of Estimated VARX Model

This example shows how to estimate a multivariate time series model that contains lagged endogenous and exogenous variables, and how to

Compare Generalized and Orthogonalized Impulse Response Functions

This example shows the differences between orthogonal and generalized impulse response functions using the three-dimensional VAR(2) model in